MESA

POLITECNICO MILANO 1863 | DIPARTIMENTO DI ELETTRONICA INFORMAZIONE E BIOINGEGNERIA

ENGINEERING CHALLENGE 2026

IT

Autonomous Driving & Secure Connected Vehicles

26/05/2026 @ PoliMi

THE SCENARIO

You just joined the PoliMi Autonomous Vehicle competition team.

To evaluate your technical skills, the team leader assigned you a challenge based on real data recorded during an autonomous racing session on a real track.

You will be evaluated on:

Technical skills / Problem solving / Work organisation / Teamwork / Quality of the final presentation / Effective demo

YOUR OBJECTIVE

Develop a perception and tracking system for autonomous racing.

The system must process multi-sensor data and answer five questions:

- **Where** are the opponent vehicles? → detection
- **How fast** are they moving? → state & motion estimation
- **Who** is who? → consistent tracking over time
- **What** does the environment look like? → live mapping
- **What next?** → trajectory & behaviour prediction

THE SENSORS YOU'LL USE

LiDAR *(Light Detection and Ranging)*

Shoots laser beams and measures how long they take to bounce back. Output: a 3D "point cloud" of the surroundings. Excellent geometry, no colour.

Radar *(Radio Detection and Ranging)*

Uses radio waves and the Doppler effect. Gives you distance AND relative velocity directly. Robust to rain, dust, low light.

Camera + Odometry

Cameras add colour and texture, great for classification. Odometry tells you the ego vehicle's own speed and motion.

ROS BAGS & THE DATASET

What is ROS?

Robot Operating System: the de-facto standard middleware for robotics. A “ROS 2 bag” is a recording of every sensor message that travelled on the robot during a run: you can replay it later as if the car were still driving.

What you receive

- **2025-10-08_09-35_sensors_raw:** ROS 2 bag with LiDAR, Radar, Cameras (see metadata.yaml)
- **ego.mat:** kinematic and inertial signals of the ego vehicle

How to read it: don’t panic *(any language, any framework allowed)*

Python: **rosbags** library (no full ROS install needed) • MATLAB: **ros2bagreader** • ROS 2 native: **ros2 bag play**

LINKS

- **Challenge Instructions and these slides**:

https://drive.google.com/drive/folders/1uZnhDTafprcdaWhB6v3ry60DUV-bZsgu?usp=drive_link

- **Data:**

https://www.dropbox.com/scl/fo/g0ramm3cecmeqq1zdhavz/AI--5Ef4fRE7HpJstvcJi4s?rlkey=exgado9ccwgizx10nuhbxlg98&st=fcb1orze&dl=0